

A University Curriculum Course for Undergraduates: Artificial Intelligence and Art

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ABSTRACT

This paper describes a new course entitled “AI & Art” offered at Quinnipiac University in the spring of 2023. An updated version of this course is scheduled for Spring 2024. In this course, students use text-to-image AI generators to create artwork, write prompts using ChatGPT, and write short essays on critical issues and topics (creativity, deep fakes, copyright etc.) implicated by the rise of AI image-generation software. This course fosters essential learning outcomes such as critical thinking, creativity, and hands-on experience with AI technology preparing students for 21st-century careers.

CCS CONCEPTS

• General and reference; • Document types; • General conference proceedings;

KEYWORDS

Artificial Intelligence, AI, text-to-image generation, AI Detection, ChatGPT

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1 INTRODUCTION

AR250*DA was first offered in the spring of 2023 as a special topics online course at Quinnipiac University [Quinnipiac University, 2023], which satisfies a breadth requirement of the required undergraduate curriculum. Students began working with Open AI's DALL·E2 [Open AI, 2023] followed by Midjourney [Midjourney, 2023], an even more powerful text-to-image generator. Students to learn how to write effective prompts to input to the AI to achieve the desired visual output. Students also experiment with ChatGPT [Open AI, 2022] to help them generate prompts. Students write short responses to critical questions such as: “What is art?”; “What is creativity?”; “How can you tell if an artwork is AI generated?”; “Should artists be afraid of AI-generated art?”; “The problem of deep fakes!”; “Is there gender or racial bias in AI image generators?” etc.

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This new course empowers graduates to successfully navigate the 21st century. Such innovative courses match the mission of higher education evolve to provide pathways that not only spark creativity, problem-solving and critical thinking, but also support students and faculty as they incubate new ideas in supportive environments, supported by a well-educated workforce. Higher education institutions are the gateways to innovation.

1.1 The University Curriculum

As a studio arts class AI & Art, satisfies one three credit, Fine Arts Course requirement as part of the State of Connecticut Department of Higher Education mandated breadth and depth requirements.

2 A LIBERAL EDUCATION

A Quinnipiac University education provides students with both specialized knowledge of a discipline, and a broad understanding of human cultures and the physical and natural world—the essence of the Liberal Education.

2.1 Essential Learning Outcomes

Students acquire Essential Learning Outcomes (ELOs) through a purposeful integration of the University Curriculum, requirements within one's major, and co-curricular experiences. This course fosters: critical and creative thinking; inquiry and analysis; effective communication; knowledge and literacies; social, emotional and intercultural intelligence.

3 ASSIGNMENTS

Students are given the following assignments during this course: weekly image generation homework; weekly readings and short essay responses; prompt engineering and prompt sharing (blog posting).

3.1 Student Work

Students write text prompts to explore genres like portraits, still life, and landscapes, while experimenting different rendering styles such as photo realistic, animé, silk-screen, painterly, impressionistic and/or expressionistic.

3.2 Using ChatGPT and AI Detection Software

Postings should be ‘likely’ written by a human, with a low probability of being AI generated. AI detectors are used for spot checks. If two or more AI detection reports indicate the text is likely written by AI students are required to rewrite their assignment.

4 INCLUSIVE EXCELLENCE

In this course, a diversity of opinions and full participation of all students are encouraged. All members of the community are asked to think about the work they are producing and ask themselves if they are consciously or unconsciously reproducing or reinforcing stereotypes, bias, or other elements that reinforce systemic racism, sexism, bigotry or other inequalities.

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